# Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

**SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT** **JEE-MAIN** **Date: 09-01-2026**
**Time: 09.00Am to 12.00Pm** **GTM-11** **Max. Marks: 300**

## KEY SHEET

## MATHEMATICS

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | **4** | 2 | **3** | 3 | **2** | 4 | **2** | 5 | **3** |
| 6 | **3** | 7 | **4** | 8 | **4** | 9 | **1** | 10 | **3** |
| 11 | **4** | 12 | **3** | 13 | **2** | 14 | **2** | 15 | **1** |
| 16 | **3** | 17 | **3** | 18 | **4** | 19 | **4** | 20 | **2** |
| 21 | **12** | 22 | **41** | 23 | **825** | 24 | **6** | 25 | **7** |

## PHYSICS

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 26 | **3** | 27 | **3** | 28 | **2** | 29 | **1** | 30 | **3** |
| 31 | **3** | 32 | **4** | 33 | **1** | 34 | **1** | 35 | **2** |
| 36 | **1** | 37 | **3** | 38 | **2** | 39 | **2** | 40 | **1** |
| 41 | **3** | 42 | **4** | 43 | **3** | 44 | **1** | 45 | **2** |
| 46 | **43** | 47 | **9** | 48 | **250** | 49 | **36** | 50 | **5** |

## CHEMISTRY

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 51 | **4** | 52 | **1** | 53 | **4** | 54 | **4** | 55 | **1** |
| 56 | **2** | 57 | **1** | 58 | **2** | 59 | **1** | 60 | **4** |
| 61 | **3** | 62 | **4** | 63 | **1** | 64 | **2** | 65 | **1** |
| 66 | **1** | 67 | **4** | 68 | **3** | 69 | **3** | 70 | **3** |
| 71 | **2** | 72 | **4** | 73 | **4** | 74 | **46** | 75 | **30** |



For More Join: @IITJEE_Advanced

# SOLUTION

# MATHEMATICS

1.

$6(a_2+a_4)(a_4+a_6)=3$

$T_6+T_8=a_1r^5+a_1r^7=2$

$\Rightarrow a_1r^5(1+r^2)=2$ .....(1)

$T_3\cdot T_5=\frac{1}{9}\Rightarrow a_1r^2\cdot a_1r^4=\frac{1}{9}\Rightarrow a_1^2r^6=\frac{1}{9}$

$\Rightarrow a_1r^3=\frac{1}{3}$ .....(2)

$\Rightarrow a_1r^3r^2(1+r^2)=2\Rightarrow\frac{1}{3}r^2(1+r^2)=2$

$\Rightarrow r^4+r^2-6=0\Rightarrow(r^2+3)(r^2-2)=0\Rightarrow r^2=2$

$a_1=\frac{1}{6\sqrt{2}}$

$6(a_2+a_4)(a_4+a_6)=6(a_1r+a_1r^3)(a_1r^3+a_1r^5)=6a_1^2r^4(1+r^2)^2=3$

2.

$2P(X=2)=3P(x=1)\Rightarrow\ 2.\ n_{c_2}p^2\cdot(1-p)^{n-2}=3\cdot{}^nC_1p(1-p)^{n-1}$

$2.\frac{n(n-1)}{2}p=3.n(1-p)$

$\Rightarrow(n-1)p=3-3p\Rightarrow(n+2)p=3\Rightarrow p=\frac{3}{n+2}$

$Mean-variance=np-npq=1\Rightarrow np^2=1\Rightarrow p^2=\frac{1}{n}$

$\Rightarrow\frac{9}{(n+2)^2}=\frac{1}{n}\Rightarrow n^2+4n+4=9n$

$\Rightarrow n^2-5n+4=0$

$\Rightarrow n=4$

$\therefore p=\frac{3}{6}=\frac{1}{2}$

$n^2p(x>1)=16(1-p(x=0)-p(x=1))$

$=16\left(1-4_{c_0}\left(\frac{1}{2}\right)^4-4_{c_1}\left(\frac{1}{2}\right)^4\right)=11$

3. Let $z=x+iy$

$\overline{z}=i(z^2+Re(\overline{z}))$

$x-iy=i(x^2-y^2+2ixy+x)$

$\Rightarrow x=-2xy$ and $x^2-y^2+x+y=0$

$\Rightarrow x=0$ or $y=-1/2$

If $x=0;y^2-y=0\Rightarrow y(y-1)=0\Rightarrow y=0,1$

If $y=-1/2;x^2+x-3/4=0\Rightarrow 4x^2+4x-3=0$

$\Rightarrow(2x-1)(2x+3)=0\Rightarrow x=-3/2,1/2$

$z=0;i,\frac{1}{2}-\frac{i}{2},\frac{-3}{2}-\frac{i}{2}$

Sum of $|z|^2=0+1+\frac{1}{2}+\frac{10}{4}=\frac{16}{4}=4$



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$$\left|\bar{a}\times\bar{b}\right| = |a||b|\sin\left(\bar{a},\bar{b}\right) = 2\times3\times\frac{1}{\sqrt{2}} = 3\sqrt{2}$$

4. $$\left|\left(\bar{a}+2\bar{b}\right)\times\left(2\bar{a}-3\bar{b}\right)\right|^2 = \left|-3\left(\bar{a}\times\bar{b}\right)+4\left(\bar{b}\times\bar{a}\right)\right|^2 = 49\,|\,\bar{b}\times\bar{a}\,|^2$$

$$= 49(3\sqrt{2})^2 = 49\times9\times2 = 882$$

5.

$$\int_0^{\frac{\pi}{4}} e^{-x}\left(\tan^{49}x+\tan^{51}x\right)dx = \int_0^{\frac{\pi}{4}} e^{-x}\tan^{49}x\left(\sec^2x\right)dx$$

$$= \left(e^{-x}\cdot\frac{\tan^{50}x}{50}\right)_0^{\frac{\pi}{4}} + \int_0^{\frac{\pi}{4}} e^{-x}\frac{\tan^{50}x}{50}dx$$

$$= \frac{e^{-\frac{\pi}{4}}}{50} + \frac{1}{50}\int_0^{\frac{\pi}{4}} e^{-x}\tan^{50}x\,dx$$

$$= \frac{e^{\frac{-\pi}{4}} + \int_0^{\frac{\pi}{4}} e^{-x}\tan^{50}x\,dx}{50}$$

$$50 = \frac{e^{\frac{-\pi}{4}} + \int_0^{\frac{\pi}{4}} e^{-x}\tan^{50}x\,dx}{\int_0^{\frac{\pi}{4}} e^{-x}\left(\tan^{49}x+\tan^{51}x\right)dx}$$

6.

$$\lim_{x\to0}\left(\frac{e^{ax}-\cos bx-\frac{cx}{2}e^{-cx}}{x^2\left(\frac{1-\cos2x}{x^2}\right)}\right)$$

$$= \frac{1}{2}\lim_{x\to0}\left(\frac{e^{ax}-\cos bx-\frac{cx}{2}e^{-cx}}{x^2}\right)\left(\frac{0}{0}\right)$$

$$= \frac{1}{2}\lim_{x\to0}\left(\frac{ae^{ax}+b\sin bx-\frac{c}{2}\left(-xce^{cx}+e^{-cx}\right)}{2x}\right)\left(\frac{0}{0}\right)$$

As $Dr\to0$ as $x=0, Nr\to0$

$$a+0-\frac{c}{2}=0 \Rightarrow 2a=c$$

Again use L' Hospitals rule and put x = 0

$$\frac{1}{2}\left(a^2+b^2+\frac{c^2}{2}(1-0+1)\right)=34 \qquad \Rightarrow a^2+b^2+c^2=68 \Rightarrow 5a^2+b^2=68$$

7.

$$x^2-\sqrt{2}x+2=0$$

$$x=\frac{\sqrt{2}\pm\sqrt{2-8}}{2}$$

$$=\frac{\sqrt{2}\pm i\sqrt{6}}{2}$$

$$=\sqrt{2}\left[\frac{1\pm i\sqrt{3}}{2}\right]$$

$$\alpha=\sqrt{2}\left[\frac{1+i\sqrt{3}}{2}\right]=-\sqrt{2}\left[\frac{-1-i\sqrt{3}}{2}\right]=-\sqrt{2}\omega^2$$

$$\beta=\sqrt{2}\left[\frac{1-i\sqrt{3}}{2}\right]=-\sqrt{2}\left[\frac{-1+i\sqrt{3}}{2}\right]=-\sqrt{2}\omega$$

$$\therefore \alpha^{14}+\beta^{14}=2^7\left[\omega^{28}+\omega^{14}\right]$$

$$2^7\left[\omega+\omega^2\right]$$

$$2^7(-1)=-128$$



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8. Verification ($4^{th}$ option)

$$\begin{vmatrix} -3+1 & 1-2 & 5-5 \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix} = \begin{vmatrix} -2 & -1 & 0 \\ -3 & 1 & 5 \\ -1 & 2 & 5 \end{vmatrix}$$

$$= -2(-5)+1(-10) = 10-10 = 0$$

9. Rank $= 2 \times 5! + 3 \times 4! + 2 \times 3! + 1 \times 2! + 0 \times 1! + 1$

$= 240 + 72 + 12 + 2 + 1$

$= 327$

$$r = \frac{np-k}{p+q}, where\, n=5, p=3, q=2, k=5$$

$$= \frac{5(3)-5}{5} = 2$$

10.

$$Now, T_{2+1} = 5C_2 \left(2x^3\right)^3 \left(\frac{-1}{3x^2}\right)^2$$

$$= \frac{10 \cdot 2^3}{3^2} x^5 = \frac{80}{9} x^5$$

$$|A| = 1(6-1)-2(2\alpha-1)+3(\alpha-3) = 5-4\alpha+2+3\alpha-9 = 2$$

$$\Rightarrow -\alpha-2 = 2 \Rightarrow \alpha = -4$$

$$\left|2adj(2adj(2A))\right| = 2^3 \left|adj(2adj(2A))\right|$$

$$= 2^3 \,|\, 2adj(2A) \,|^2$$

11. $$= 2^3 \left(2^3\right)^2 \left| adj(2A) \right|^2$$

$$= 2^9 \left|2A\right|^4 = 2^9 \left(2^3\right)^4 |A|^4$$

$$= 2^9 2^{12} 2^4 = 2^{25}$$

$$= \left(2^5\right)^5 = 32^5, \quad n = 5$$

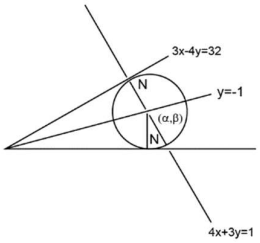
$$3n + \alpha = 15 - 4 = 11$$

$$\left.\begin{aligned} 2x+y-z &= 5 \\ 2x-5y+\lambda z &= \mu \\ x+2y-5z &= 7 \end{aligned}\right\} \Rightarrow 6y-(\lambda+1)z = 5-\mu \;\&\; y-3z = 3$$

12. $$\frac{6}{1} = \frac{\lambda+1}{3} = \frac{5-\mu}{3}$$

$$\lambda+1 = 18 \Rightarrow \lambda = 17, 18 = 5-\mu \Rightarrow \mu = -13$$

$$(\lambda+\mu)^2 + (\lambda-\mu)^2 = 4^2 + 30^2 = 916$$




13. Equation of angular bisectors of the tangents is



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$$\frac{3x+4y-24}{5}=\pm\left(\frac{3x-4y-32}{5}\right)$$

$\Rightarrow 3x+4y-24=3x-4y-32$

$\Rightarrow 8y=-8\Rightarrow y=-1$

$centre(\alpha,\beta)lies\ on\ y=-1\Rightarrow \beta=-1$

$\sin ce\ 4x+3y=1 is\ normal$

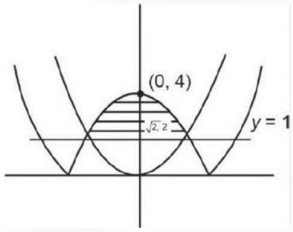
$\Rightarrow 4\alpha+3\beta=1\Rightarrow 4\alpha-3=1$

$\Rightarrow \alpha=1(\alpha,\beta)=(1,-1)$

$r=\frac{|3-4-24|}{5}=5$

$\therefore \alpha-\beta+r=1+1+5=7$

14.




$=2\left(\int_1^2\sqrt{y}dy+\int_2^4\sqrt{4-y}dy\right)$

$=\frac{4}{3}\left(\left(2\sqrt{2}-1\right)-\left(0-2\sqrt{2}\right)\right)=\frac{4}{3}\left(4\sqrt{2}-1\right)$

15. Equation of plane:

$$\begin{vmatrix} x & y+1 & z-2 \\ 1 & -3 & 1 \\ 2 & 1 & -3 \end{vmatrix}=0$$

$\Rightarrow x(8)-(y+1)(-5)+(z-2)(7)=0$

$\Rightarrow 8x+5y+5+7z-14=0$

$\Rightarrow 8x+5y+7z=9$

Passes through (-2, 5, 0)

16. Points of intersection of the given lines are A(6, 8), B(1, 2)& C(5, -7)

Centroid of

$\Delta ABC=\left(\frac{6+1+5}{3},\frac{8+2-7}{3}\right)$

$(\alpha,\beta)=(4,1)$

$\alpha+2\beta=4+2=6$

$2\alpha-\beta=8-1=7$

$x^2 - 13x + 42 = 0$

17. Dr 's of $\overrightarrow{AB}$ = (3, 12, 3)

i.e., (1, 4, 1)

Equation of AB: $\frac{x-4}{1}=\frac{y-5}{4}=\frac{z-8}{1}=\lambda(say)$



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$N=(\lambda+4,4\lambda+5,\lambda+8)$

$Dr's\ of \overrightarrow{PN}=(\lambda+3,4\lambda+7,\lambda+5)$

$PN\perp AB\Rightarrow 1(\lambda+3)+4(4\lambda+7)+1(\lambda+5)=0$

$\Rightarrow 18\lambda+36\Rightarrow\lambda=-2$

$N=(2,-3,6)$

$\perp r$ Distance form N to 2x - 2y + z + 5 = 0 is 7.

18. $\frac{x^2}{x^2+1}\geq 0$ also $x^2<x^2+1\Rightarrow\frac{x^2}{x^2+1}<1$

Range of

$f(x)=\left[4\sin^{-1}(0),4\sin^{-1}(1)\right)$

$=[0,4\pi/2)=[0,2\pi)$

19. $\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CA}=\overline{0}\Rightarrow -2\hat{i}+\hat{j}+3\hat{j}-\alpha\hat{i}-\beta\hat{j}-r\hat{k}+4\hat{i}+3\hat{j}+\overline{0k}=\overline{0}$

$\Rightarrow(-2-\alpha+4)\hat{i}+(1-\beta+3)\hat{j}+(3-r+\delta)=0$

$\Rightarrow -2-\alpha+4=0;4-\beta=0;3-r+\delta=0$

$\Rightarrow\alpha=2;\beta=4;r=\delta+3$

$\Delta=\frac{1}{2}\left|\overrightarrow{AB}\times\overrightarrow{AC}\right|=\frac{1}{2}\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\-2&1&3\\4&3&\delta\end{vmatrix}=5\sqrt{6}$

$\Rightarrow\left|\hat{i}(\delta-9)-\hat{j}(-2\delta-12)+\hat{k}(-6-4)=10\sqrt{6}\right|$

$\Rightarrow\delta^2-18\delta+81+4(\delta^2+12\delta+36)=500$

$\Rightarrow 5\delta^2+30\delta-275=0\Rightarrow\delta^2+6\delta-55=0$

$\Rightarrow(\delta^2+11)(\delta-5)=0\Rightarrow\delta=5$

$\overrightarrow{CA}\cdot\overrightarrow{CB}=60$

20. $x^2+y^2-4x-16y+64=0$

Centre (2, 8) radius = 2

shortest distance will lie along common normal

normal at P(t) to parabola is

$\therefore y+tx=2t+t3$

Since it will pass through (2, 8) $\Rightarrow 8+2t=2t+t^3\ \ t=2,\ p(t)\equiv p(t^2,2t)\equiv P(4,4)$

$\therefore d=\sqrt{4+16}=\sqrt{20}\therefore d^2=20$

21. $S\&S'=(\pm 2,0),e=\frac{3}{2},\ 2ae=4\Rightarrow ae=2\Rightarrow a\cdot\frac{3}{2}=2\Rightarrow a=\frac{4}{3}\ b^2=a^2e^2-a^2\Rightarrow b^2=4-\frac{16}{9}=\frac{20}{9}$

Tangent is 3x - 2y + k = 0

$y=\frac{3x}{2}+\frac{k}{2}$

$c^2=a^2m^2-b^2\Rightarrow\frac{16}{9}\times\frac{9}{4}-\frac{20}{9}=\frac{k^2}{4}$

$\Rightarrow\frac{144-80}{9\times 4}=\frac{k^2}{4}\Rightarrow k=\pm\frac{8}{3}$

$3x-2y\pm\frac{8}{3}=0$

$3x-2y=\frac{8}{3}\Rightarrow\frac{x}{8/9}=\frac{y}{-8/6}=1$

$|6a|+|5b|=6\cdot\frac{8}{9}+5\cdot\frac{8}{6}=12$



For More Join: @IITJEE_Advanced

22. $f_n=\int_0^{\pi/2}\left(\sum_{k=1}^{n}\sin^{k-1}x\right)\left(\sum_{k=1}^{n}(2k-1)\sin^{k-1}x\right)\cos x\,dx$

put sinx=t then cosx dx=dt

$$f_n=\int_0^1\left(\sum_{k=1}^{n}(t)^{k-1}\right)\left(\sum_{k=1}^{n}(2k-1)(t)^{k-1}\right)dt$$

$$f_{21}-f_{20}=\int_0^1\left[\left(\sum_{k=1}^{21}(t)^{k-1}\right)\left(\sum_{k=1}^{21}(2k-1)(t)^{k-1}\right)-\left(\sum_{k=1}^{20}(t)^{k-1}\right)\left(\sum_{k=1}^{20}(2k-1)(t)^{k-1}\right)\right]dt$$

$$=\int_0^1\left[\left(1+t+t^2+....+t^{19}\right)(41)t^{20}+\left(1+3t+5t^2+....+41t^{20}\right)t^{20}\right]dt$$

$$=\left(\frac{1}{21}+\frac{1}{22}+....+\frac{1}{40}\right)41+\left(\frac{1}{21}+\frac{3}{22}+.....+\frac{39}{40}+\frac{41}{41}\right)$$

$$=\left[\frac{42}{21}+\frac{44}{22}+\frac{46}{23}+.....+\frac{80}{40}+\frac{41}{41}\right]=40+1=41$$

23.

$$\left[\sqrt{1}\right]+\left[\sqrt{2}\right]+\left[\sqrt{3}\right]+....+\left[\sqrt{120}\right]$$

$$1+1+1+2+2+2+2+2+3+3+3+3+3+3+3+...$$

$$=1(3)+2(5)+3(7)+4(9)+...+10(21)$$

$$=\sum_{r=1}^{10}r(2r+1)=\sum_{r=1}^{10}\left(2r^2+r\right)$$

$$=2\sum_{r=1}^{10}r^2+\sum_{r=1}^{10}r$$

$$=2\frac{10\cdot11\cdot21}{6}+\frac{10\cdot11}{2}=825$$

$$\text{I.F}=e^{2\int\frac{2x}{x^2-1}dx}=e^{2\log\left(x^2-1\right)}=\left(x^2-1\right)^2$$

$$\text{Sol. } y\left(x^2-1\right)^2=\int\frac{x+2}{\left(x^2-1\right)^{5/2}}\left(x^2-1\right)^2dx$$

$$=\int\frac{x+2}{\sqrt{x^2-1}}dx=\frac{1}{2}\int\frac{2x}{\sqrt{x^2+1}}dx+2\int\frac{dx}{\sqrt{x^2-1}}$$

$$y\left(x^2-1\right)^2=\sqrt{x^2-1}+2\log\left|x+\sqrt{x^2-1}\right|+C$$

$$\text{Put } x=2, y=\frac{2}{9}\left(\log 2+\sqrt{3}\right)$$

$$\Rightarrow\frac{2}{9}\log_e\left(2+\sqrt{3}\right)(4-1)^2=\sqrt{4-1}+2\log\left|2+\sqrt{3}\right|+C$$

$$\Rightarrow C=-\sqrt{3}\Rightarrow y\left(x^2-1\right)^2=\sqrt{x^2-1}+2\log\left|x+\sqrt{x^2-1}\right|-\sqrt{3}$$

$$\text{put } x=\sqrt{2}$$

$$y=1+2\log\left|\sqrt{2}+1\right|-\sqrt{3}$$

24. $y=2\log\left|\sqrt{2}+1\right|+1-\sqrt{3}$

Here $\alpha=2,\beta=1,\gamma=3$

$\Rightarrow\alpha\beta\gamma=6$

25. A={−4,−3,−2,0,1,3,4}

R=(a,b)∈A×A:b=|a| or b2=a+1}

R={(−4,4)(−3,3)(0,0)(1,1)(3,3)(4,4)(3,−2)(0,1)}

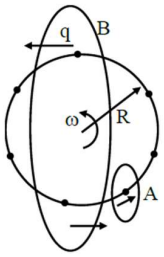
Required elements (-4, 4)(-3,- 3)(-2, -2) (4, -4)(3, -3)(1, 0)(-2, 3)= 7



For More Join: @IITJEE_Advanced

# PHYSICS

26.




$$I_A = \frac{Nq}{\frac{2\pi}{\omega}}$$

$$I_A = \frac{Nq\omega}{2\pi}, I_B = 0$$

$$I_A = I_B = \frac{Nq\omega}{2\pi}$$

27. $hv = \phi + KE_{max}$

$KE_{max} = eV_0$

$$V_0 = \frac{hv - \phi}{e}$$

(A)$V_0$v/sV is linear correct

(B) Slope

$$v_0 = \left(\frac{h}{e}\right)v - \frac{\phi}{e}$$ Wrong

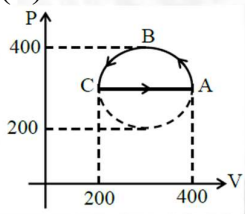
Slope $\frac{h}{e}$

(C) Correct

(D) Incorrect

(E) Correct

28.

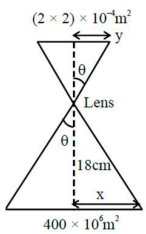



$$W = \frac{1}{2}\pi R^2$$

$$= \frac{1}{2} \times \pi \times \left(\frac{200}{2} \times 10^3\right) \times \frac{200}{2} \times 10^{-6}$$

$$= \frac{10\pi}{2} = 5\pi J$$

29.






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$A_{image} = 400km^2$

$x = 20\times10^3 m = 2\times10^4 m$

$y = 2\times10^{-2} m$

$\frac{x}{y} = 10^6 = \frac{18Km}{f}$

$f = 18\times10^{-3} m = 18mm$

$f = 1.8cm$

$A_{image} = 400km^2$

$x = 20\times10^3 m = 2\times10^4 m$

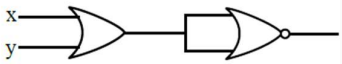
$y = 2\times10^{-2} m$

$\frac{x}{y} = 10^6 = \frac{18Km}{f}$

$f = 18\times10^{-3} m = 18mm$

$f = 1.8cm$

30.




| x | y | |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

$\frac{T_2 - T_1}{t} = K\left[T_{avg} - T_s\right]$

$T_1 = 24^\circ C; T_2 = 40^\circ C, t = 4, T_s = 16^\circ C$

$\frac{40-24}{4} = K[32-16]$

$K = \frac{4}{16} = \frac{1}{4}$

31. Now $\frac{24-T}{4} = K\left[\frac{T+24}{2} - 16\right]$

$24 - T = \frac{T-16}{2} + 16$

$\frac{3T}{2} = 28$

$T = \frac{56}{3} C$

$[E] = M^1L^2T^{-2}$

$[Pc] = M^1L^1T^{-1} \cdot L^1T^{-1} = M^1L^2T^{-2}$

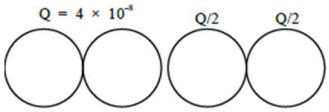
32. $\left[mc^2\right] = M^1L^2T^{-2}$

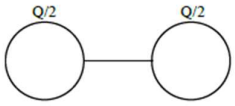
$E^2 = M^1L^2T^{-2}$

$E^2 = P^2c^2 + m^2c^4$



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33.

$$F=\frac{k\left(\frac{Q}{2}\right)\left(\frac{Q}{2}\right)}{r^2}$$

$$9\times10^{-3}=\frac{9\times10^{9}\times\left(4\times10^{-8}\right)\times4\times10^{-8}}{4\times r^2}$$

$$r^2=\frac{9\times10^{9}\times16\times10^{-16}}{4\times9\times10^{-3}}=4\times10^{-4}$$

$$r=2\times10^{-2}\,m=2\,cm$$

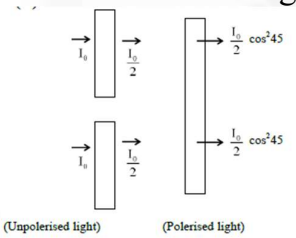
$$x=x_0\sin^2\frac{t}{2}=x_0\left(\frac{1-\cos t}{2}\right)$$

$$x-\frac{x_0}{2}=\frac{-\cos t}{2}$$

34. where $x_0=1$

$$x-\frac{1}{2}=\frac{-\cos t}{2}$$

Particle is oscillating between x = 0 to x = 1




35.

after passing through third poleriser, Intensity of both the waves must be $\frac{I_0}{4}$

now, at a point where path diff is $\frac{\lambda}{3}$, phase difference

$$\Delta\phi=2K\left(\frac{\Delta x}{\lambda}\right)=\frac{2\pi}{3}$$

$$\therefore I_{res}=\sqrt{\left(\frac{I_0}{4}\right)^2+\left(\frac{I_0}{4}\right)^2+2\left(\frac{I_0}{4}\right)^2\cos\frac{2\pi}{3}}$$

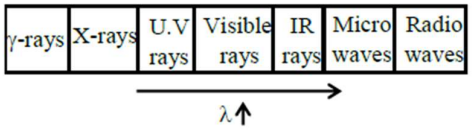
$$=\frac{I_0}{4}$$

36. $$\beta=\left(\frac{\lambda_0}{\mu}\right)\times\frac{D}{d}=\frac{690\times10^{-9}\times0.72}{1.44\times1.5\times10^{-3}}=0.23mm$$



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37.




$\bar{F} = q(\bar{V} \times \bar{B})$

38. $\bar{F} = 0$

$\bar{V} \| \bar{B}$

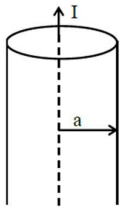
$\theta = 0 \text{or} 180$

$$\frac{\text{Linear KE}}{\text{RotationalK.E}} = \frac{\frac{1}{2}mv_{cm}^2}{\frac{1}{2}I\omega^2}$$

39. $$\frac{mv_{cm}^2}{\frac{2}{5}mR^2\omega^2} = \frac{5}{2}$$

$(V = \omega R)$

40.




$$B_{in} = \frac{\mu_0(\bar{J} \times \bar{r})}{2}$$

$B_{in} \propto r$

$$B_{out} = \frac{\mu_0 I}{2\pi r}$$

$$B_{net} \propto \frac{1}{r}$$

41. $(A) T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$

Temp increases

R) Free expansion is assumed fast, so Adiabatic

$$U_1 = \frac{4Kq_0^2}{a} + \frac{2Kq_0^2}{\sqrt{2}a} = \frac{Kq_0^2}{a}(4+\sqrt{2})$$

42. $$U_2 = \frac{Kq_0^2}{\left(\frac{a}{\sqrt{2}}\right)}(4+\sqrt{2}) = \frac{Kq_0^2}{a}(4\sqrt{2}+2)$$

$$U_2 - U_1 = \frac{Kq_0^2}{a}(3\sqrt{2}-2)$$

$\vec{r} = 5t^2\hat{i} - 5t\,\hat{j}$

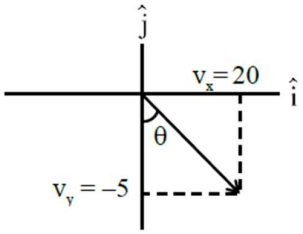
43. $\vec{v} = 10t\hat{i} - 5\hat{j}$

$\vec{v} = 20\hat{i} - 5\hat{j} \quad \text{at} = 2\,\text{sec}$

$\Rightarrow |\vec{v}| = \sqrt{400+25} = 5\sqrt{17}\ \text{m/s}$



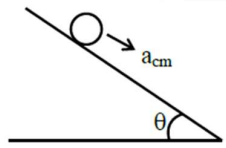
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$$\tan\theta = \frac{20}{5} = 4$$

$$\theta = \tan^{-1} 4$$

From-veY-axis

44. 


$$t = \sqrt{\frac{2\ell}{a_{cm}}}$$

$$a_{cm} = \frac{g\sin\theta}{1+\frac{I_{cm}}{MR^2}}$$

$$a_{cm} = \frac{g\sin\theta}{1+\frac{I_{cm}}{MR^2}}$$

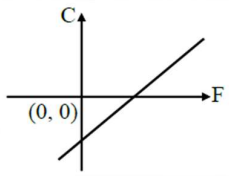
$$a_1 = a_{cm_1} = \frac{5g\sin\theta}{7} \text{.....Solid}$$

$$a_2 = a_{cm_2} = \frac{3g\sin\theta}{5} \text{.....Hollow}$$

$$a_1 > a_2$$

$$t_1 < t_2$$

45. $\frac{C}{5} = \frac{F-32}{9} \Rightarrow C = \frac{5F}{9} - \frac{160}{9}$




46. Since bulk modulus is given as

$$B = \frac{-\Delta P}{\left(\frac{\Delta V}{V}\right)}$$

$$2.15\times10^9 = \frac{-\Delta P}{-\left(\frac{0.2}{100}\right)}$$

$$\Delta P = 2.15\times10^9 \times 2\times10^{-3}$$

$$= 4.3\times10^6 = 43\times10^5 \text{ N/m}^2$$



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47. ∵ acceleration due to gravity on surface is given by
Now since diameter is reduced to 1/3rd, radius also reduces to 1/3rd, keeping mass constant New value of acceleration due to gravity on Earth's surface is

$$g' = \frac{GM}{\left(\frac{R_e}{3}\right)^2} = 9\frac{GMe}{R_e^2} = 9g$$

48. Since time period of a revolving charge is $\frac{2\pi m}{qB}$

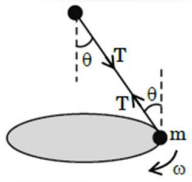
Where B = magnetic field, due to a solenoid = $\mu_0 nI$

$$\therefore T = \frac{2\pi m}{q(\mu_0 nI)}$$

$$75\times10^{-9} = \frac{(2\pi)(9\times10^{-31})}{1.6\times10^{-19}\times4\pi\times10^{-7}\times n\times1.5}$$

$$N = 250$$

49.




$$T\cos\theta = mg.....(1)$$

$$T\sin\theta = M\omega^2 R.....(2)$$

Using equation (2)

$$T\sin\theta = M\omega^2(L\sin\theta)$$

$$T = M\omega^2 L = M\left(\frac{3}{\pi}\times2\pi\right)^2 L$$

$$T = 36ML$$

50. Since power emitting by a source is given as $= \frac{\text{Total energy emitted}}{\text{time}}$

$$= \frac{(E_1\text{photon})\times\text{Number of photons}(N)}{t} \quad P_1 = (E_1)n$$

$$\frac{P_1}{P_2} = \frac{(E_1)n_1}{(E_2)n_2} = \frac{\left(\frac{hC}{\lambda_1}\right)n_1}{\left(\frac{hC}{\lambda_2}\right)n_2} \quad \frac{P_1}{P_2} = \left(\frac{\lambda_2}{\lambda_1}\right)\frac{n_1}{n_2}$$

Substituting the given values

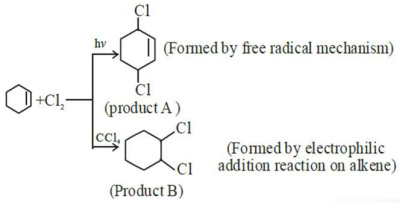
$$2 = \left(\frac{300}{600}\right)\times\frac{2\times10^{15}}{n_2}$$

$$n_2 = \frac{1}{2}\times10^{15} = 5\times10^{14}\ \text{Photon/sec}$$



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# CHEMISTRY

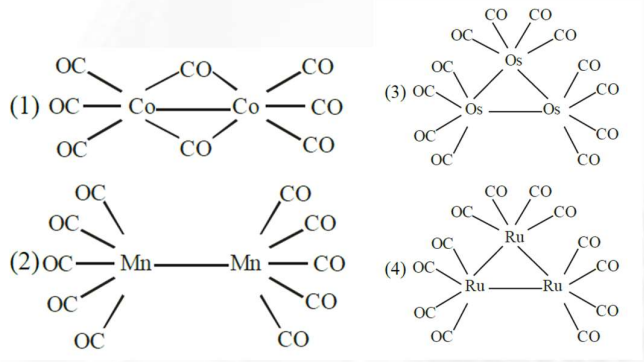
51. 


Hence correct Ans. (4)

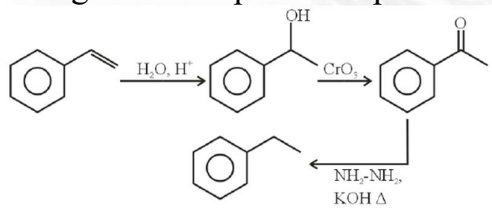
52. (1) Fluorspar is $CaF_2$

53. It is a type of conjugation responsible for resonance.

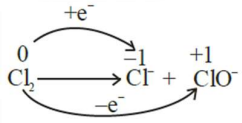
54. $2KMnO_4 \xrightarrow{\Delta} K_2MnO_4 + MnO_2 + O_2$

55. 


56. Proteins are natural polymers composed of α-amino acids which are connected by peptide linkages. Hence proteins upon acidic hydrolysis produce α-amino acids.

57. 


58. $(\Delta G)_{P,T}$=(+) ve for non-spontaneous process

59. 


$\Rightarrow Cl_2 + 2\bar{O}H \rightarrow Cl^- + ClO^- + H_2O$

60. $2MnO_4^- + H_2O + I^- \xrightarrow{\text{alkaline medium}} 2MnO_2 + 2OH^- + IO_3^-$

61. First ionisation energy increases along the period. Along the period Z increases which outweighs the shielding effect.

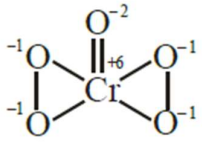
62. Ziegler catalyst → Titanium
Blood pigment → Iron
Wilkinson catalyst → Rhodium
Vitamin $B_{12}$→Cobalt

$Cl^- + K_2Cr_2O_7 + H_2SO_4 \rightarrow CrO_2Cl_2 \xrightarrow{\text{Basic medium}} \underset{\text{yellow solution}}{CrO_4^{2-}} + Cl^-$

63. $\underset{\text{yellow solution}}{CrO_4^{2-}} \xrightarrow[3.10\%H_2O_2]{\text{1.Acidification 2. Amyl alcol}} \underset{\text{blue compound}}{CrO_5}$



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64. The difference in energy between the actual structure and the lowest energy resonance structure for the given compound is known as resonance energy.

65. Gr-14 EN
C 2.5
Si 1.8
Ge 1.8
Sn 1.8
Pb 1.9
The electronegativity values for elements from Si to Pb are almost same. So Statement I is false.

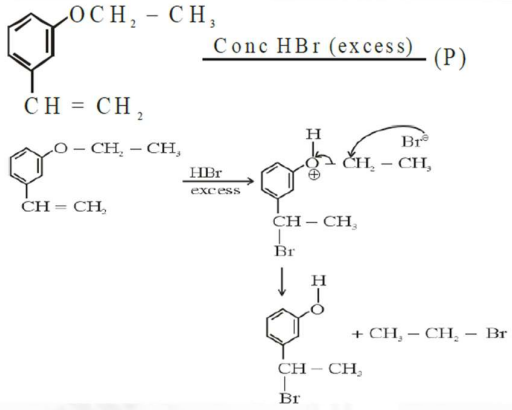
$Rb = [Kr]5s^1$

$n = 5$

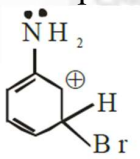
66. $l = 0$

$m = 0$

$s = +1/2 \text{ or } -1/2$

67. 

68. Since $-NH_2$ group is o/p directing hence arenium ion will not be formed by attack at meta position i.e.



Hence Answer is (3)

69. $Fe^{2+} \xrightarrow[\text{Conc.}H_2SO_4]{H^+} Fe^{+3}$

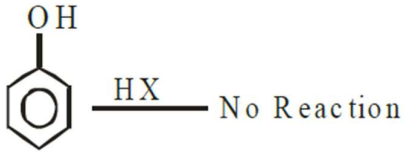
$Fe^{+3} \xrightarrow{^-SCN} Fe(SCN)_3 \text{ (blood red colour)}$

Appearance of blood red colour indicates presence of both nitrogen and sulphur.

70. Assertion (A): Given statement is correct because in phenol hydroxyl group cannot be replaced by halogen atom.
Reason (R) :



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Given reason is false

Hence Assertion (A) is correct but Reason (R) is false

$K_P = K_C \cdot (RT)^{\Delta n_g}$

71. $\Delta n_g = 1$

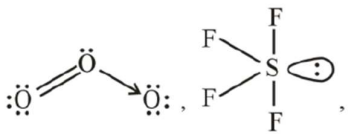
$\Rightarrow K_c = \frac{K_p}{RT} = \frac{0.492}{0.082 \times 300} = 2 \times 10^{-2}$

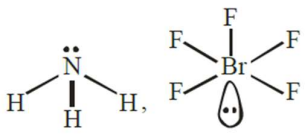
$M = \frac{n_{solute}}{V} \times 1000$

72. $= \frac{\left(\frac{31.4}{98}\right)}{\left(\frac{100}{1.25}\right)} \times 1000$

$= 4.005 \approx 4$







73.

$Zn^{+2} + 2e^- \rightarrow Zn$

$W = Z \times i \times t$

74. $= \frac{65.4}{2 \times 96500} \times 0.015 \times 15 \times 60$

$= 45.75 \times 10^{-4}\,gm$

$K = \frac{K_1 \cdot K_2}{K_3} = \frac{A_1 \cdot A_2}{A_3} \cdot e^{\frac{(E_{a_1} + E_{a_2} - E_{a_3})}{RT}}$

75. $A \cdot e^{-E_a/RT} = \frac{A_1 A_2}{A_3} \cdot e^{-\frac{(E_{a_1} + E_{a_2} - E_{a_3})}{RT}}$

$E_a = E_{a_1} + E_{a_2} - E_{a_3} = 40 + 50 - 60 = 30 kJ/mole.$



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